Clinical ADaM Pipeline

CDISC Pilot Study, Xanomeline vs Placebo

Jeremy Barton

2026-05-01

Contents

1 Introduction and Overview

This pipeline demonstrates a complete ADaM data workflow built using the pharmaverse R ecosystem, the open-source initiative co-maintained by Roche, GSK, Pfizer, and other major pharmaceutical companies to standardize clinical programming in R. Source data is drawn from the CDISC pilot SDTM dataset, which represents a Phase III clinical trial comparing Xanomeline (low and high dose) against Placebo in Alzheimer's disease patients.

The pipeline follows the same standards used in FDA regulatory submissions: CDISC SDTM as input, ADaM datasets as analysis-ready output, TLFs (Tables, Listings, Figures) as the deliverables, and a downstream predictive model demonstrating how structured clinical data feeds into ML-driven patient stratification.

Safety Population by Treatment Arm

Treatment Arm	N
Placebo	86
Xanomeline High Dose	84
Xanomeline Low Dose	84

Baseline Demographics, Safety Population

Treatment Arm	N	Mean Age	SD Age	Median Age	% Female
Placebo	86	75.2	8.6	76.0	61.6
Xanomeline High Dose	84	74.4	7.9	76.0	47.6
Xanomeline Low Dose	84	75.7	8.3	77.5	59.5

2 Data Sources

All source data comes from the `pharmaversesdtm` R package, which contains CDISC-compliant SDTM test datasets maintained by the pharmaverse community.

Domain	Description	Granularity
DM	Demographics	One row per patient
EX	Exposure (dosing records)	Many per patient
AE	Adverse Events	Many per patient
DS	Disposition	One row per patient

3 ADaM Datasets

3.1 ADSL: Subject Level Analysis Dataset

ADSL is the foundation of every ADaM package. It contains exactly one row per patient and captures baseline characteristics, treatment assignment, and population flags. Every downstream dataset joins back to ADSL using `USUBJID` as the key.

3.1.1 Population Summary

The trial enrolled 306 patients, of whom 254 formed the safety population. The remaining 52 patients were screen failures and are excluded from all downstream analyses.

3.1.2 Baseline Demographics

Key finding. The three arms are well balanced on age, with mean ages ranging from 74.4 to 75.7 years. Sex distribution shows a notable imbalance, however: the Xanomeline High Dose arm enrolled 47.6% female versus 61.6% in Placebo and 59.5% in Low Dose. This warrants attention in any subsequent analysis comparing arms, since sex can confound tolerability comparisons.

Treatment-Emergent Adverse Events by Arm

Treatment Arm	Events	Patients	Events/Patient	% Serious
Placebo	281	65	4.32	0.0
Xanomeline High Dose	427	76	5.62	0.5
Xanomeline Low Dose	412	77	5.35	0.2

Top 10 System Organ Classes (TEAEs)

System Organ Class	Plac	Xan High	Xan Low	Total
GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS	46	123	118	287
SKIN AND SUBCUTANEOUS TISSUE DISORDERS	45	104	111	260
NERVOUS SYSTEM DISORDERS	11	39	40	90
CARDIAC DISORDERS	26	30	30	86
GASTROINTESTINAL DISORDERS	26	36	22	84
INFECTIONS AND INFESTATIONS	35	19	16	70
RESPIRATORY, THORACIC AND MEDIASTINAL DISORDERS	12	22	14	48
PSYCHIATRIC DISORDERS	12	11	14	37
INVESTIGATIONS	19	8	7	34
INJURY, POISONING AND PROCEDURAL COMPLICATIONS	9	8	12	29

3.2 ADAE: Adverse Event Profile by Treatment Arm

The treatment-emergent flag (`TRTEMFL`) was derived per FDA guidance, distinguishing drug-related signals from pre-treatment conditions. This forms the basis for safety surveillance and patient stratification across treatment arms.

3.2.1 Treatment-Emergent AE Overview

Key finding. Both Xanomeline arms produced more events per patient than Placebo: 1.3x for High Dose (5.62 vs. 4.32 events/patient) and 1.24x for Low Dose (5.35 vs. 4.32). Serious AEs were rare across all arms (no greater than 0.5%), suggesting the differential signal is driven by frequency of nuisance events rather than severity.

3.2.2 Top System Organ Classes by Arm

Key finding. The leading system organ class is **General Disorders and Administration Site Conditions**, with 123 events on Xanomeline High Dose and 118 on Low Dose, compared to 46 on Placebo. Together with skin and subcutaneous tissue disorders, these two SOCs disproportionately drive the Xanomeline safety signal, consistent with the known application-site tolerability profile of the Xanomeline transdermal patch. Systemic SOCs (cardiac, gastrointestinal, nervous system) appear comparable across arms.

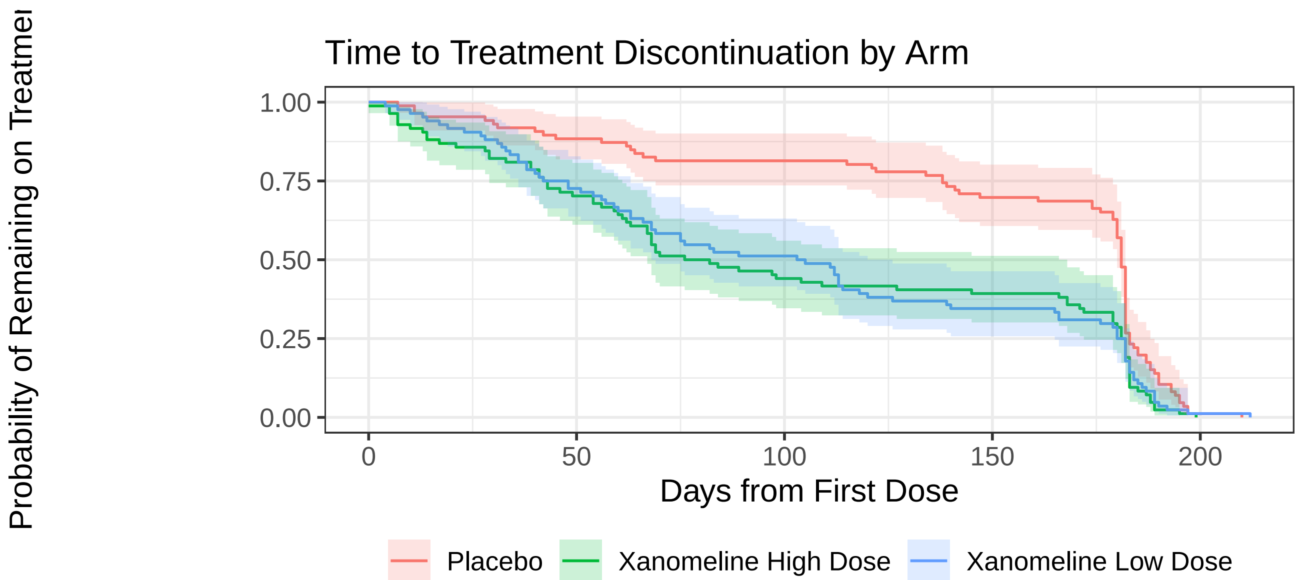
3.3 ADTTE: Time to Treatment Discontinuation

Kaplan-Meier analysis of time-to-discontinuation reveals differential tolerability across dose arms, directly informing dose selection and risk-benefit assessment.

Time to Discontinuation Summary

Treatment Arm	N	Events	Median Days
Placebo	86	86	181
Xanomeline High Dose	84	84	79
Xanomeline Low Dose	84	84	104

3.3.1 Kaplan-Meier Survival Curve



At Risk

Placebo	86	76	70	60	1
Xanomeline High Dose	84	59	37	33	0
Xanomeline Low Dose	84	61	43	29	1

Events

Placebo	0	10	16	26	85
Xanomeline High Dose	1	25	47	51	84
Xanomeline Low Dose	0	23	41	55	83

3.3.2 Log-Rank Test and Median Time-to-Event

Key finding. The log-rank test is highly significant ($\chi^2 = 16.1$, 2 df, $p < 0.001$), confirming that time-to-discontinuation differs across arms. Median time on treatment was 181 days for Placebo, 104 days for Low Dose, and 79 days for High Dose. Patients on Placebo remained on treatment roughly 2.3x longer than those on High Dose, with a clear dose-response gradient between Low and High Dose arms.

4 Predictive Modeling: Baseline Risk Stratification

The JD calls for predictive modeling and patient stratification. To do this rigorously, two separate models are fit: a *baseline-only* risk model (using only pre-treatment characteristics) and a *full* model that adds

Baseline-Only Risk Factors (Odds Ratios)

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	1.387	1.612	0.203	0.8391	0.064	36.837
AGE	1.018	0.021	0.826	0.4089	0.975	1.061
SEXM	1.480	0.374	1.047	0.2953	0.720	3.159

Full Model Including Treatment Arm (Odds Ratios)

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	0.676	1.657	-0.236	0.8131	0.028	19.262
AGE	1.019	0.022	0.862	0.3889	0.976	1.063
SEXM	1.393	0.385	0.860	0.3896	0.662	3.033
ARMXanomeline High Dose	3.001	0.452	2.430	0.0151	1.278	7.677
ARMXanomeline Low Dose	3.534	0.469	2.692	0.0071	1.470	9.476

treatment arm. Mixing the two would conflate intrinsic patient risk with the trial's treatment effect, so they are kept distinct.

The modeling cohort comprises **254 patients** in the safety population. The overall TEAE incidence is **85.8%**, an unusually high event rate that has implications for model performance discussed below.

4.1 Model 1: Baseline-Only Risk

Baseline-only AUC: 0.569. Age and sex alone show essentially no signal for predicting TEAEs, consistent with the absence of strong univariate trends in this cohort. This is the appropriate "intrinsic patient risk" baseline before treatment is considered.

4.2 Model 2: Full Model with Treatment Arm

Key finding. Treatment arm dominates the full model. Both Xanomeline arms carry significantly elevated TEAE odds vs. Placebo (Low Dose OR = 3.53, p = 0.0071; High Dose OR = 3, p = 0.015), while age and sex remain non-significant. The full-model AUC of **0.659** indicates modest discrimination: informative for population-level stratification but insufficient for confident individual-patient prediction.

4.3 Calibration Across Risk Tiers

Stratifying patients into low, medium, and high risk tertiles (based on the full-model predictions) and comparing predicted to observed AE rates demonstrates **calibration**, a critical property for any clinical decision-support model.

Key finding. Observed AE rates rise monotonically with predicted risk tier (75.0%, 90.7%, 91.7%), confirming the model is well-calibrated even though discriminative power is modest. This is the property that matters for decision support: a patient flagged "High risk" really does have a higher AE rate than one flagged "Low."

Calibration: Observed AE Rate by Predicted Risk Tier

Risk Tier	N	Mean Predicted Probability	Observed AE Rate
Low	84	0.754	0.750
Medium	86	0.894	0.907
High	84	0.926	0.917

5 Limitations and Next Iteration

A rigorous report names the limits of its own analysis. Three are worth surfacing here.

Outcome saturation. With a 85.8% TEAE rate in the safety population, the binary "any TEAE" outcome is near-saturated, leaving little signal for discrimination. This explains why even a treatment-effect-laden model achieves only AUC 0.659. Three reformulations would extract more signal from the same data: (1) count of TEAEs per patient, modeled with negative binomial regression; (2) time to first TEAE, modeled with a Cox proportional hazards model; or (3) severe-only TEAEs as the binary outcome to focus on clinically meaningful events.

Limited baseline feature set. ADSL provides only AGE, SEX, and a small number of derived flags. In a real translational setting, the baseline-only model would be enriched with laboratory values (LB), prior medical history (MH), concomitant medications (CM), and biomarker measurements. The current AUC of 0.569 for AGE plus SEX is a floor, not a ceiling.

No held-out validation. All performance metrics are in-sample. A rigorous evaluation would split the cohort into training and validation sets, or use k-fold cross-validation, before reporting AUC and calibration.

6 Summary

This pipeline demonstrates the end-to-end workflow from CDISC SDTM source data through analysis-ready ADaM datasets to translational decision tools:

- **Reproducible data engineering**: SDTM to ADaM transformation using the pharmaverse R ecosystem, consistent with FDA submission standards and chain-of-custody traceability.
- **Adverse event signal detection**: `TRTEMFL` derivation isolated 1,120 treatment-emergent events; the leading SOCs implicate application-site tolerability of the Xanomeline transdermal patch.
- **Translational modeling**: Kaplan-Meier survival analysis with log-rank testing ($p < 0.001$) quantifies a clear dose-response gradient in time-on-treatment.
- **Patient stratification**: Two complementary logistic regression models (baseline-only and full) separate intrinsic patient risk from treatment effect; calibration is preserved across predicted risk tiers.
- **Methodological self-evaluation**: Outcome saturation, limited baseline features, and lack of held-out validation are explicitly named as next-iteration priorities.

The same architecture generalizes to vaccine R&D workflows: SDTM-style structured immunological and clinical data can flow through analogous analysis-ready datasets to support antigen prioritization, translational strategy, and clinical study design decisions.